

IX. Appendices

Appendix A - Classification Class Mapping and Split Manifest Summary

This appendix consolidates the class mapping and split identity that support the classification methodology and results. It records data-partition provenance and retraining status without reproducing the performance tables from Chapter VI.

Table A.1. SDI-4 four-class mapping and fixed seed-42 split

Serialized class	Train	Validation	Test	Total
Healthy	282	60	61	403
BG	139	30	29	198
WSSV	229	49	50	328
WSSV_BG	154	33	33	220
Total	804	172	173	1,149

The primary SDI-4 split was generated once with seed 42 and preserved for directly compared classification runs. Class order remained Healthy, BG, WSSV, and WSSV_BG across manifests, training class counts, output decoding, metric scripts, and mobile metadata. Historical candidate screening inspected test-partition Macro-F1, so the test partition was not completely independent of model-family selection.

Table A.2. Additional-source mapping and retraining status

Regime	Source mapping / construction	Retained test size	Evaluation status
EXT-3-Original	Tom_BT → Healthy; Den_Mang → BG; Dom_Trang → WSSV. No WSSV_BG category.	47	The mapped three-class model was retrained in this regime; this is not direct inference by the original four-class check-point.
SDI-4 + EXT-3-Original	Each source was split independently under the nominal 70/15/15 policy, then train+train, validation+validation, and test+test were merged. WSSV_BG is supplied only by SDI-4.	220	The four-class model was retrained on the integrated partitions. Source composition and class-prior changes remain explicit interpretation factors.

The integrated regime is therefore a partition-wise retraining experiment rather than zero-shot external inference. Legacy labels such as ShrimpDB-3 or Combined-4 are retained only as historical aliases; the primary dataset identifiers used throughout the thesis are EXT-3-Original and SDI-4 + EXT-3-Original.